# L1 python™ XII

| | |
|---|---|
| 1 | Write a Python program to accept the value of an integer N and display all the factors of N.<br><br>Repeat the above for multiple integers using a user-controlled loop. |
| 2 | Write a Python program to accept the value of an integer N and display the numbers from 1 to N alongwith their cumulative sum with previous number. |
| 3 | Write a Python program to accept the values of two numbers in X and N (X as float and N as int), find the result of X raised to the power N using a loop.<br><br>Repeat the above for multiple pairs of X and N using a user-controlled loop. |
| 4 | Write a Python program to find the first 5 armstrong numbers. |
| 5 | Write a Python program to find the **sum** of first N prime numbers.<br><br>The value of N to be entered by the user. |
| 6 | Write a Python program to find the **sum** of Composite numbers between 1 to N.<br><br>The value of N to be entered by the user. |
| 7 | Write a Python program to print the first N number of fibonacci series of numbers starting from 0 and N.<br><br>The value of N to be entered by the user.<br><br>**Enter N: 7**<br>**First 7 Fibonacci Series**<br>**0 1 1 2 3 5 8** |
| 8 | Write a Python menu driven program to print the following sequence of numbers and their sum.<br><br>The value of N to be entered by the user.<br>**A. 5,10,15,... Upto Nth Term**<br>**B. 1,6,11,... Upto Nth Term**<br>**C. 10,30,50, ... Upto the Nth Term** |
| 9 | Write a Python menu driven program to print the following sum of sequence of numbers (N to be entered by user)<br><br>**A. 2+4+6+... Upto Nth Term**<br>**B. 5+11+17+... Upto Nth Term**<br>**C. 50+150+250, ... Upto the Nth Term** |
| 10 | Write a Python menu driven program to print the following sum of sequence of numbers (X and N to be entered by user) |

| | |
|---|---|
| | **A. $X + X^2 + X^3+,...$ Upto Nth Term**<br>**B. $X + X^2/2! - X^3/3!+...$ Upto Nth Term**<br>**C. $X + X^3/3! + X^5/5!, ...$ Upto the Nth Term** |

**General Instructions:**

i. Type the solution of all the problems using **Python** Language on **Python IDLE** or **Colab**

ii. Type the following on top of each of your program as comment lines

```
"""
Program List : <Practical Assignment List number appears here>
Program No   : <Question number in the list>
Developed By : <Your name appears here>
Date         : <Date when the program was coded appears here>
"""
```

iii. On successful execution with meaningful data, **copy** and **paste** the sample output at the bottom of the program as comment lines.

iv. Add all the codes in separate pages of a single Google Doc named as in the format as **<Class>-<Section>-<L1>-<Name>** For example **XII-E-L1-Namya Giri (If the Name is Namya Giri of XII-E)**

**NOTE:** Recommended **Font, "Courier New", Style "Bold" Size "12" for all programs with single line spacing. Default indentation 2 Character only.**

v. **"Turn in"** the Doc as a single document in response to this assignment submission with all programs. Once verified by the teacher, take out a hardcopy from the printer and get them signed by the respective computer teacher.

```
# ------------------------------------------------------------#
# List-Program No    : L1-P1
# Developed By       : PUJA DEY
# Date               : 14-April-2025
# ------------------------------------------------------------#
N=int(input("Enter an integer: "))
print("Factors of N are: ", end="")
for i in range(1,N+1):
 if N%i==0:
  print(i, end=",")
```

```
Enter an integer: 25
Factors of N are: 1,5,25,
```

```
while True:
  N=int(input("Enter an integer: "))
  print("Factors of N are: ", end="")
  for i in range(1,N+1):
    if N%i==0:
      print(i, end=",")
  print( )
  ch=input("Want for more numbers? (Y/N): ")
  if ch=="N":
    break
```

```
Enter an integer: 25
Factors of N are: 1,5,25,
Want for more numbers? (Y/N): Y
Enter an integer: 36
Factors of N are: 1,2,3,4,6,9,12,18,36,
Want for more numbers? (Y/N): N
```

```
# ------------------------------------------------------------#
# List-Program No    : L1-P2
# Developed By       : PUJA DEY
# Date               : 14-April-2025
# ------------------------------------------------------------#
s=0
N=int(input("Enter a no: "))
print("Number","Cumulative Sum", sep="\t")
for i in range (1,N+1):
  s+=i
  print(i,s, sep="\t" )
```

```
Enter a no: 7
Number      Cumulative Sum
1           1
2           3
3           6
4           10
5           15
6           21
7           28
```

```
# ------------------------------------------------------------#
# List-Program No    : L1-P3
# Developed By       : PUJA DEY
# Date               : 14-April-2025
# ------------------------------------------------------------#
while True:
  P=1
  X=float(input("Enter base no: "))
  N=int(input("Enter the power: "))
  for i in range(1,N+1):
    P=P*X
  print(X,"^",N, "=", P)
  ch=input("Want for more no? (Y/N): ")
  if ch=="N":
    break
```

```
Enter base no: 5
Enter the power: 3
5.0 ^ 3 = 125.0
Want for more no? (Y/N): Y
Enter base no: 25
Enter the power: 2
25.0 ^ 2 = 625.0
Want for more no? (Y/N): N
```

```
# ------------------------------------------------------------#
# List-Program No    : L1-P4
# Developed By       : PUJA DEY
# Date               : 14-April-2025
# ------------------------------------------------------------#
count = 0
num = 0
print("The first 5 Armstrong numbers are:")
while count<5:
    N=num
    dn=len(str(num))
    s=0
    while N>0:
        d=N%10
        s+=d**dn
        N//=10
    if s== num:
        print(num)
        count += 1
    num += 1
```

```
The first 5 Armstrong numbers are:
0
1
2
3
4
```

```
# ------------------------------------------------------------#
# List-Program No    : L1-P5
# Developed By       : PUJA DEY
# Date               : 14-April-2025
# ------------------------------------------------------------#
N = int(input("Enter the value of N: "))
count = 0
num = 2
s = 0
while count<N:
    for i in range(2, num):
        if num%i == 0:
            break
    else:
        s+= num
        count+= 1
    num+= 1

print("Sum of first", N, "prime numbers is:", s)
```

```
Enter the value of N: 5
Sum of first 5 prime numbers is: 28
```

```
# ------------------------------------------------------------#
# List-Program No    : L1-P6
# Developed By       : PUJA DEY
# Date               : 14-April-2025
# ------------------------------------------------------------#

N = int(input("Enter a number: "))
s= 0
for i in range(4, N + 1):
    for j in range(2, i):
        if i % j == 0:
            s+= i
            break

print("Sum of composite numbers from 1 to", N, "is:", s)
```

```
Enter a number: 11
Sum of composite numbers from 1 to 11 is: 37
```

```
# ------------------------------------------------------------#
# List-Program No    : L1-P7
# Developed By       : PUJA DEY
# Date               : 14-April-2025
# ------------------------------------------------------------#

N=int(input("Enter N: "))
A=0;B=1
print("First", N, "Fibonacci Series")
for I in range(1,N+1):
  print(A,end=" ")
  A,B=B,A+B
```

```
Enter N: 7
First 7 Fibonacci Series
0 1 1 2 3 5 8
```

```
# ------------------------------------------------------------#
# List-Program No    : L1-P8
# Developed By       : PUJA DEY
# Date               : 14-April-2025
# ------------------------------------------------------------#
s=0
t=0
u=0
print("A. 5,10,15,... Upto Nth Term")
print("B. 1,6,11,... Upto Nth Term")
print("C. 10,30,50, ... Upto the Nth Term")
ch=input("Enter your choice (A/B/C): ")
N=int(input("Enter N: "))
if ch=="A":
  for i in range(1,N+1):
    t+=5*i
    print(5*i, end=",")
  print( "=", t)
elif ch=="B":
  for i in range(0,N):
    u+=5*i+1
    print(5*i+1, end=",")
  print("=", u)
elif ch=="C":
  for i in range(1,N+1):
    s+=10*(2*i-1)
    print(10*(2*i-1), end=",")
  print( "=", s)
else:
  print("Wrong Choice")
```

```
A. 5,10,15,... Upto Nth Term
B. 1,6,11,... Upto Nth Term
C. 10,30,50, ... Upto the Nth Term
Enter your choice (A/B/C): C
Enter N: 3
10,30,50,= 90
```

```
# ------------------------------------------------------------#
# List-Program No    : L1-P9
# Developed By       : PUJA DEY
# Date               : 14-April-2025
# ------------------------------------------------------------#
s=0
t=0
u=0
print("A. 2+4+6+... Upto Nth Term")
print("B. 5+11+17+... Upto Nth Term")
print("C. 50+150+250, ... Upto the Nth Term")
ch=input("Enter your choice (A/B/C): ")
N=int(input("Enter N: "))
if ch=="A":
  for i in range(1,N+1):
    t+=2*i
    print(2*i, end=",")
  print( "=", t)
elif ch=="B":
  for i in range(1,N+1):
    u+=5*i+(i-1)
    print(5*i+(i-1), end=",")
  print("=", u)
elif ch=="C":
  for i in range(1,N+1):
    s+=50*(2*i-1)
    print(50*(2*i-1), end=",")
  print( "=", s)
else:
  print("Wrong Choice")
```

```
A. 2+4+6+... Upto Nth Term
B. 5+11+17+... Upto Nth Term
C. 50+150+250, ... Upto the Nth Term
Enter your choice (A/B/C): B
Enter N: 3
5,11,17,= 33
```

```
# ------------------------------------------------------------#
# List-Program No    : L1-P10
# Developed By       : PUJA DEY
# Date               : 14-April-2025
# ------------------------------------------------------------#

print("A. X + X^2 + X^3+,... Upto Nth Term")
print("B. X + X^2/2! - X^3/3!+... Upto Nth Term")
print("C. X + X^3/3! + X^5/5!, ... Upto the Nth Term")
ch=input("Enter your choice (A/B/C): ")
N=int(input("Enter N: "))
X=int(input("Enter X: "))
if ch=="A":
  s=0
  for i in range(1,N+1):
    s+=X**i
  print(s)

elif ch=="B":
  s=0
  for i in range(1,N+1):
    f=1
    for z in range(1,i+1):
      f*=z
    s+=(X**i)/f
  print(s)

elif ch=="C":
  s=0
  for i in range(1,2*N+1,2):
    f=1
    for z in range(1,i+1):
      f*=z
    s+=(X**i)/f
  print(s)

else:
  print("Wrong Choice")
```

```
A. X + X^2 + X^3+,... Upto Nth Term
B. X + X^2/2! - X^3/3!+... Upto Nth Term
C. X + X^3/3! + X^5/5!, ... Upto the Nth Term
Enter your choice (A/B/C): C
Enter N: 3
Enter X: 10
1010.0
```